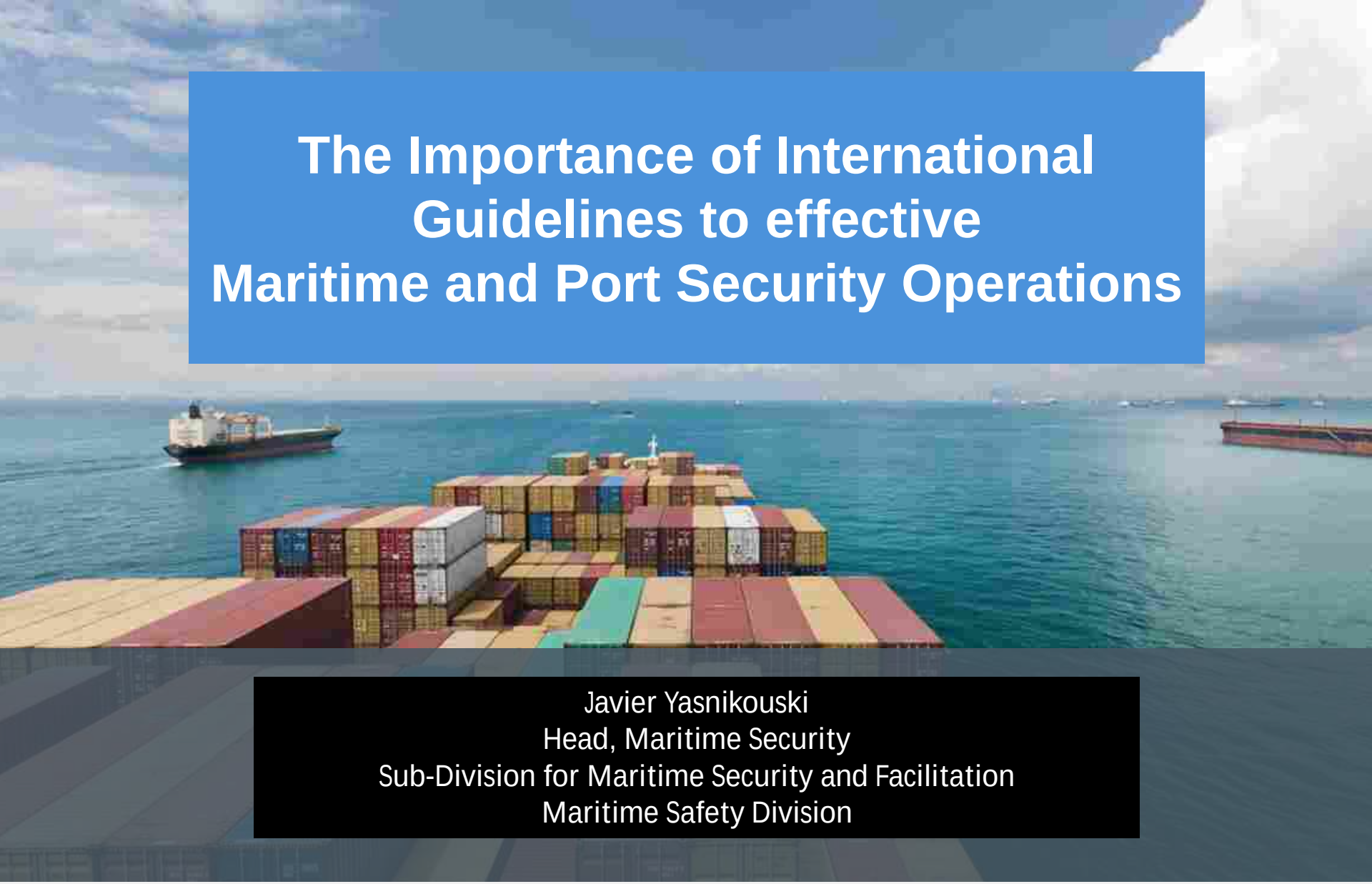
The Importance of International Guidelines to effective Maritime and Port Security Operations



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Sub-Division for Maritime Security and Facilitation
Maritime Safety Division

Contents of this presentation

- International Maritime Organization (IMO) - Overview
- IMO's efforts to enhance maritime security
- Maritime security legislation
- Maritime cyber risk management
- Current and future electronic developments
- Piracy and armed robbery against ships
- World Maritime Day 2017
- IMO's technical cooperation programme

IMO – the International Maritime Organization

IMO mission:

Safe, secure and efficient shipping on clean oceans







IMO – the International Maritime Organization

Ten largest contributors to IMO in 2015. Assessed contributions based on flat base rate with additional components based on ability to pay and merchant fleet tonnage

Panama	£5.22m	17.33%
Liberia	£3.00m	9.98%
Marshall Is.	£2.41m	8.01%
Singapore	£1.83m	6.06%
Bahamas	£1.31m	4.35%
UK	£1.30m	4.29%
Malta	£1.29m	4.27%
China	£1.20m	3.98%
Japan	£1.02m	3.39%
Greece	£1.02m	3.38%

IMO – Structure

- **Assembly**
  171 Member Governments
- **Council**
  40 Member Governments
  - **Maritime Safety**
    - SHIP DESIGN AND CONSTRUCTION (SDC)
    - SHIP SYSTEMS AND EQUIPMENT (SSE)
    - HUMAN ELEMENT, TRAINING AND WATCHKEEPING (HTW)
    - NAVIGATION, COMMUNICATION AND SEARCH AND RESCUE (NCSR)
    - CARRIAGE OF CARGOES AND CONTAINERS (CCC)
    - IMPLEMENTATION OF IMO INSTRUMENTS (III)
  - **Technical Cooperation**
  - **Facilitation**
  - **Legal**
  - **Marine Environment Protection**
    - POLLUTION PREVENTION AND RESPONSE (PPR)





IMO – Progress of measures at IMO

Casualty/ Review/ Technology → **Proposal to IMO** → Discuss, agree to refer on → **Draft text** → **Adoption or approval**



No Enforcement Role

The main focus of the IMO is to develop and adopt international instruments or regulations for safe, secure and efficient shipping on clean oceans

The implementation and enforcement of these instruments is the responsibility of Member States

IMO has no enforcement role



Impetus for Action

Safety……

- Titanic catastrophe led to international safety regulations



Environment……

- Torrey Canyon grounding led to MARPOL Convention



Security……

- Hijacked aircrafts 11 September 2001

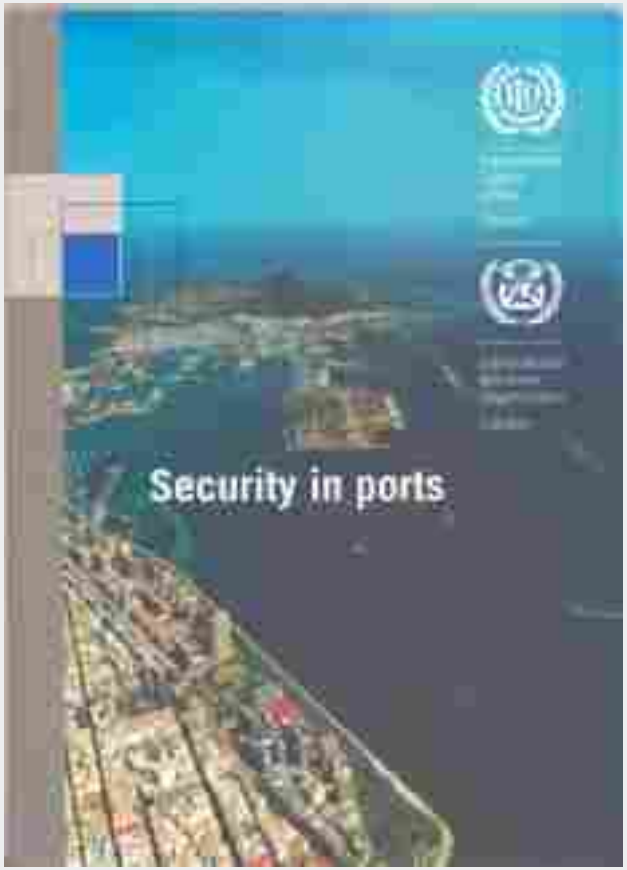


London Diplomatic Conference



– December 2002

Conference Resolution 8



Enhancement of security in cooperation with the International Labour Organization

- Guidance on wider port security
- Recommends development of port security strategy and appointment of port security committee

Conference Resolution 9

Enhancement of security in cooperation with the World Customs Organization

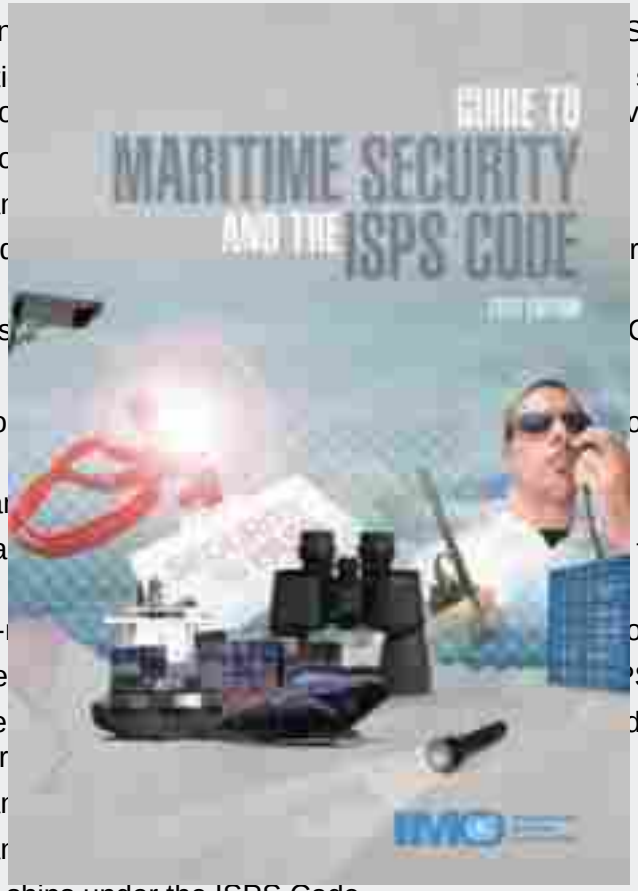
- IMO requested WCO to urgently consider measures to enhance security throughout international movements of closed CTUs
- WCO adopted in June 2005 the SAFE Framework of Standards



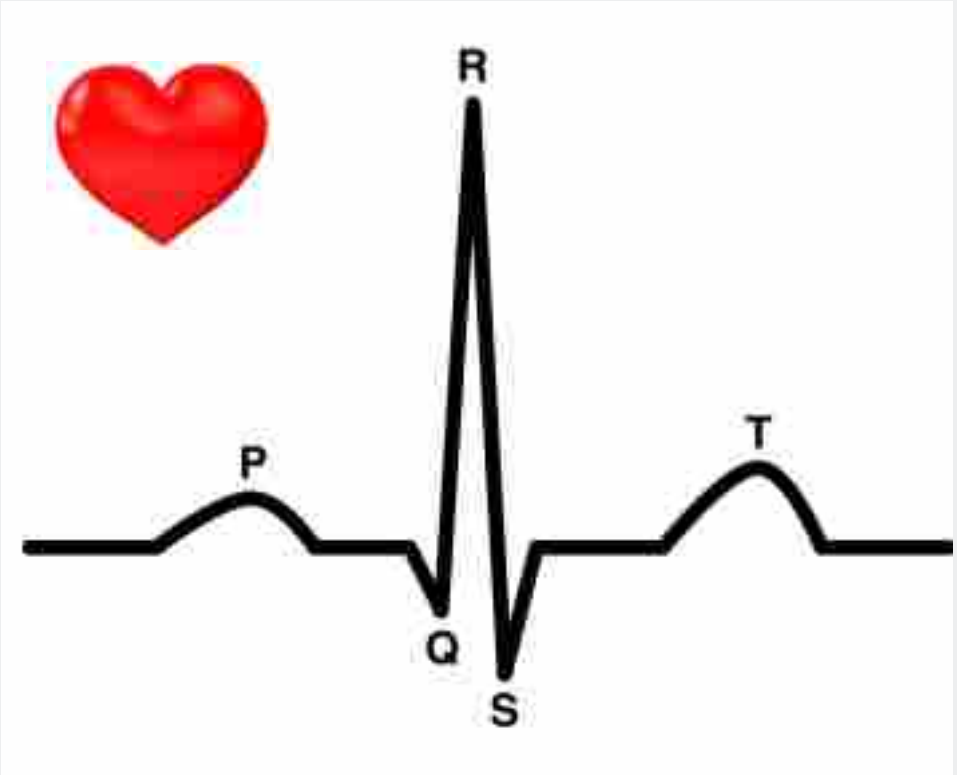
Related guidance

- Significant related guidance produced to date:

- Amendments to SOLAS chapter XI-2 and the ISPS Code;
- Amendments to 1978 STCW Convention Security Officers;
- Amendments to 1978 STCW Conventi shipboard personnel with and without security-related duties, adopted as part c vention and STCW Code;
- Guidelines on training and certification fc
- Guidelines on security-related training ar
- Performance standards, guidelines and rity alert systems (SSAS) and related matters;
- Guidance on voluntary self-assessments Companies, port facilities and ships;
- Interim guidance on the authorization of
- Guidance on the access of public autho onboard ships to which SOLAS chapter XI-2 and the ISPS Code apply;
- Interim guidance on control and compliar
- Non-mandatory guidelines on security a fall within the scope of SOLAS chapter XI-2 and the ISPS Code;
- Guidance on the submission of security- o port;
- Guidelines for the implementation of spe S Code;
- Guidance on the implementation of the d the ISPS Code, in the context of the World Customs Organization's SAFE Fr
- Guidelines on security-related training ar
- Guidelines on security-related training ar
- Guidance on shore leave and access to ships under the ISPS Code.

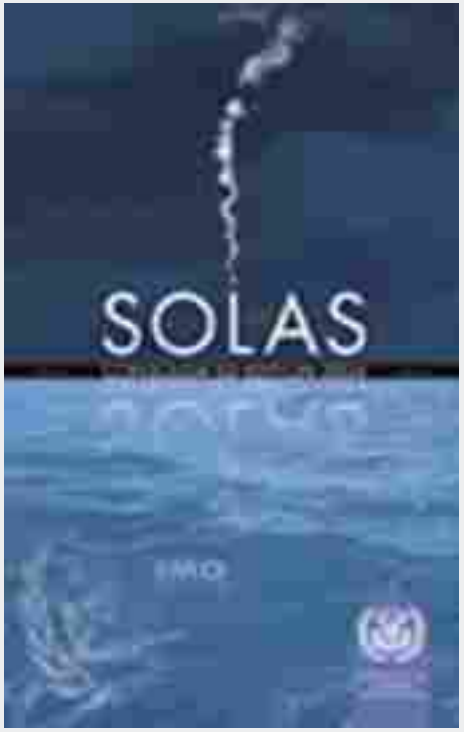


Implementation of the ISPS Code



Level of emphasis

Maritime Security Legislation







Guidance for the Development of National Maritime Security Legislation

The Maritime Safety Committee, at its 96th session (11 to 20 May 2016) approved the Guidance for the development of national maritime security legislation (MSC.1/Circ.1525).

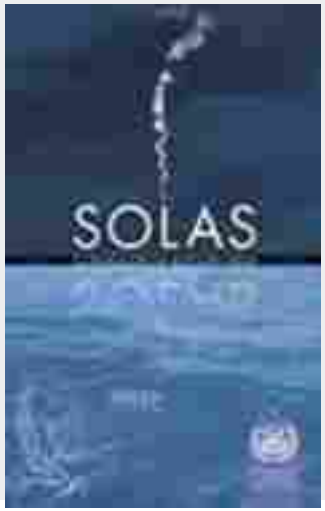




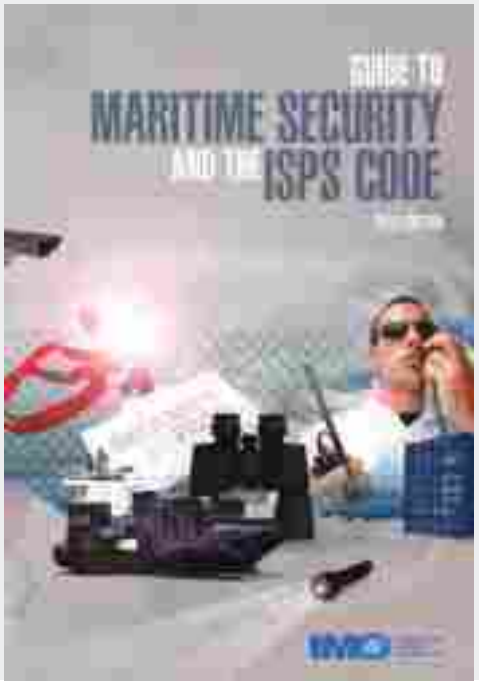
Guidance for the Development of National Maritime Security Legislation

The Guidance is drawn from the following resources:

- the SOLAS Convention
- parts A and B of the ISPS Code
- the 2012 IMO "Guide to Maritime Security and the ISPS Code", and
- related IMO resolutions and circulars.







SOLAS chapter XI-2 and the ISPS Code in force for 13 years

Many Governments **are still striving to implement** fully the maritime security measures, particularly those pertaining to port facilities



Reasons include:

- the **lack of legal and policy instruments** required to achieve compliance with the ISPS Code
- **jurisdictional issues** between Government agencies (partly as a result hereof)





Guidance for the Development of National Maritime Security Legislation

MSC.1/Circ.1525 of May 2016:

- **The drafting and enactment of appropriate national legislation** is essential to the successful implementation and oversight of the maritime security measures.

- The legislation should **specify the powers needed for Government officials to undertake their duties**, including
  - inspection and testing of security measures and procedures
  - application of enforcement actions to correct incidents of non-compliance.



Guidance for the Development of National Maritime Security Legislation

Legislation should cover:

.1 General provisions (definitions, application, etc.)

.2 National authorities for ship and port security (duties, powers, enforcement, inspections, audits, etc.)

.3 Specific provisions for ship and port facility security (Designated Authority and Administration;

.4 Enforcement and control measures.

Voluntary nature

- While the compilation in the new MSC circular provides the framework to assist in the development of national legislation, it is not intended as an auditing or assessment tool.

- Further, while this Guidance refers to mandatory provisions from the SOLAS Convention, as well as both mandatory provisions and voluntary guidance in the ISPS Code*, the use of this Guidance is not mandatory.*

Recommended Provisions

National Maritime Security Committee

A national maritime security committee should be formed to address the development, relevance and acceptability of a national maritime security framework or strategy.

A national maritime security committee should involve representatives of those regulated: major stakeholders in the port and shipping industries, port workers and seafarers, and cargo and passenger interests.



Recommended Provisions

Examples:

RSO Oversight
Security Plans
Security Level Coordination
Company, ship and port facility security officers
Training, drills and exercises
Communications
Cargo operations and ship's stores
Security Incidents
Best Management Practices
Administrative and criminal enforcement

Recommended Provisions

Part 5 on **Enforcement** represents a new focus, particularly with a view to ensuring that **national legislation is in place**:

Essential to the successful implementation and oversight of the ISPS Code is the drafting and enactment of appropriate national legislation to provide for the full implementation and oversight of the maritime security measures. The legislation should ***specify the powers needed for government officials to undertake the application of enforcement actions to correct incidents of non-compliance.***



Cyber Security



Maritime cyber risk management

MSC.1/Circ.1526 on Interim guidelines on maritime cyber risk management: Approved by MSC 96 in May 2016 due to the urgent need to raise awareness on cyber risk threats and vulnerabilities.

Provide high-level recommendations on maritime cyber risk management to safeguard shipping from current and emerging cyberthreats and vulnerabilities and include functional elements that support effective cyber risk management.



Maritime cyber risk

Maritime cyber risk refers to a measure of the extent to which a technology asset is threatened by a potential circumstance or event, which may result in shipping-related operational, safety or security failures as a consequence of information or systems being corrupted, lost or compromised.

Maritime

Cyber

Risk



Maritime cyber risk management

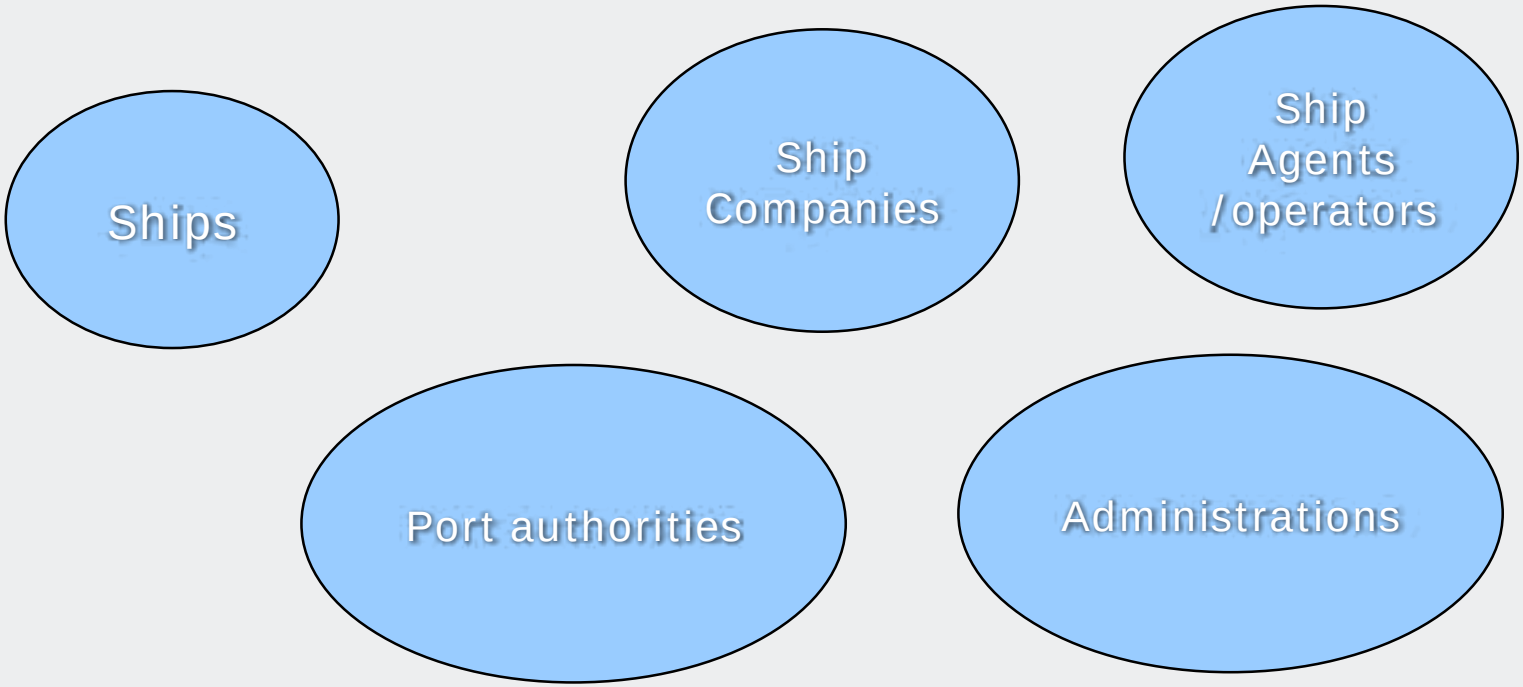
Cyber risk management means the process of identifying, analysing, assessing and communicating a cyber-related risk and accepting, avoiding, transferring or mitigating it to an acceptable level, considering costs and benefits of actions taken to stakeholders.

The Overall goal is to support safe and secure shipping, which is operationally resilient to cyber risks.



Maritime cyber risk management

The Guidelines are primarily intended for all organizations in the shipping industry, and are designed to encourage safety and security management practices in the cyber domain.



Maritime cyber risk management



Who is involved?

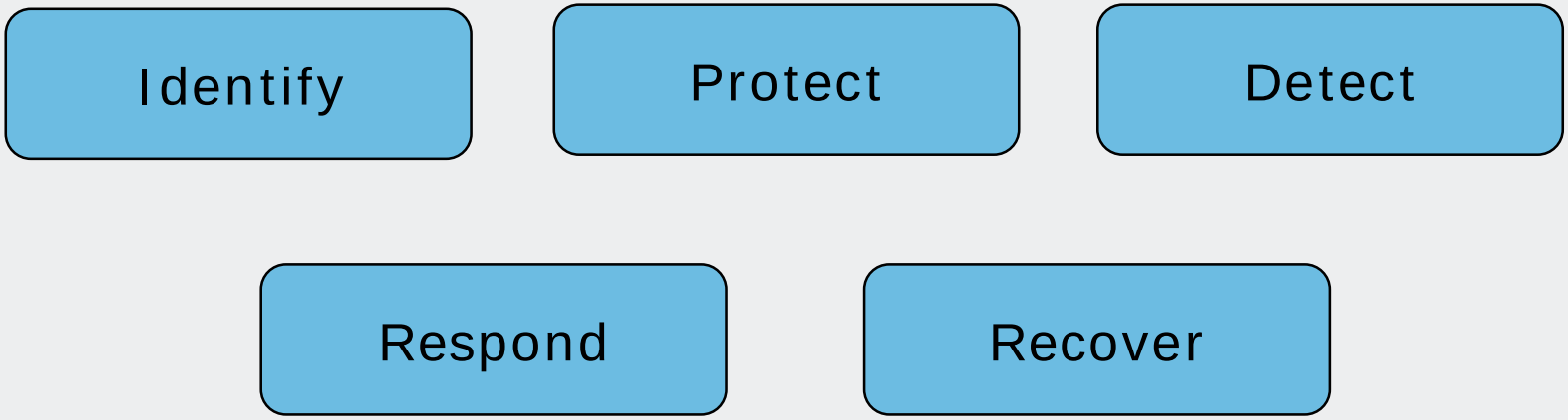
Everybody should be involved (crew members, passengers, shipping companies, etc.). However, effective cyber risk management should start at the senior management level.

A culture of cyber risk awareness and discipline should be embedded into all levels of an organization. The level of awareness and preparedness should be appropriate to roles and responsibilities in the cyber risk management system.

A holistic and flexible cyber risk management regime should be in continuous operation and constantly evaluated through effective feedback mechanisms.

Maritime cyber risk management

Functional elements to support effective cyber risk management:



Maritime cyber risk management

To address the rapidly evolving technologies and changing threats, the Guidelines recommend a risk management approach to cyber risks that is resilient and evolves as a natural extension of existing safety and security management practices established by this Organization.





Maritime cyber risk management

Additional guidance and standards may include:

- The Guidelines on Cyber Security on board Ships by BIMCO, CLIA, ICS, INTERCARGO and INTERTANKO.
- ISO/IEC 27001 standard on Information technology – Security techniques – Information security management systems – Requirements. Published jointly by the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC).
- United States National Institute of Standards and Technology's Framework for Improving Critical Infrastructure Security (the NIST Framework).

Current and future electronic developments





- Ship reporting
- E-navigation
- Maritime Single Window
- Ship communications
- AIS > VDEs
- ECDIS
- Electronic data exchange
- Access to internet

Vulnerable systems onboard ships



- Bridge systems
- Cargo handling and management systems
- Propulsion and machinery management and power control systems
- Access control systems
- Passenger servicing and management systems
- Passenger facing public networks
- Administrative and crew welfare systems and
- Communication systems

Software quality assurance

MSC.1/Circ.1512 on Guideline on Software Quality Assurance and Human-Centred Design for e-navigation: Approved by MSC 95 in June 2015.

SQA focuses on defining and testing software quality and how that helps meet user requirements to ensure that high quality, robust, testable and stable software is used in e-navigation systems.

The basic premise of HCD is that systems are designed to suit the characteristics of intended users and the tasks they perform, rather than requiring users to adapt to a system.



Latest FAL amendments

Electron

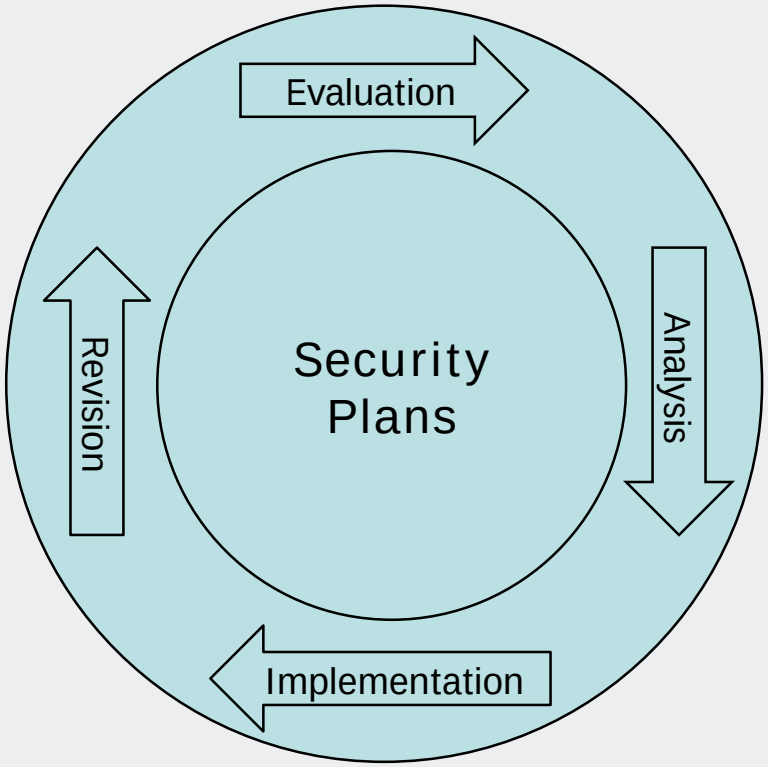


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What has changed since 2004?

- More than **11,500 port facilities** with approved Port Security Plans.
- Better coordination of security measures between ships and port facilities.
- Enhanced awareness of the importance of effective implementation of security measures and the continuous improvement.

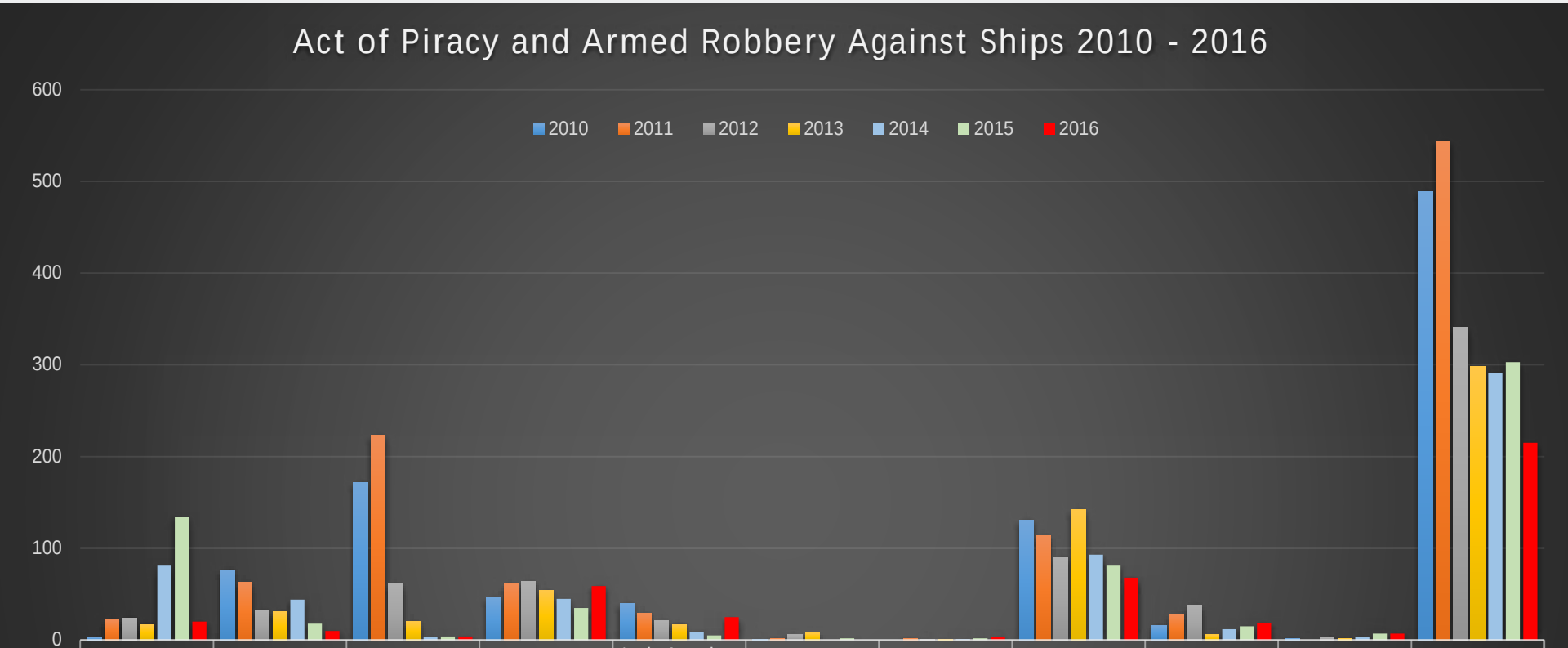


What has changed since 2004?



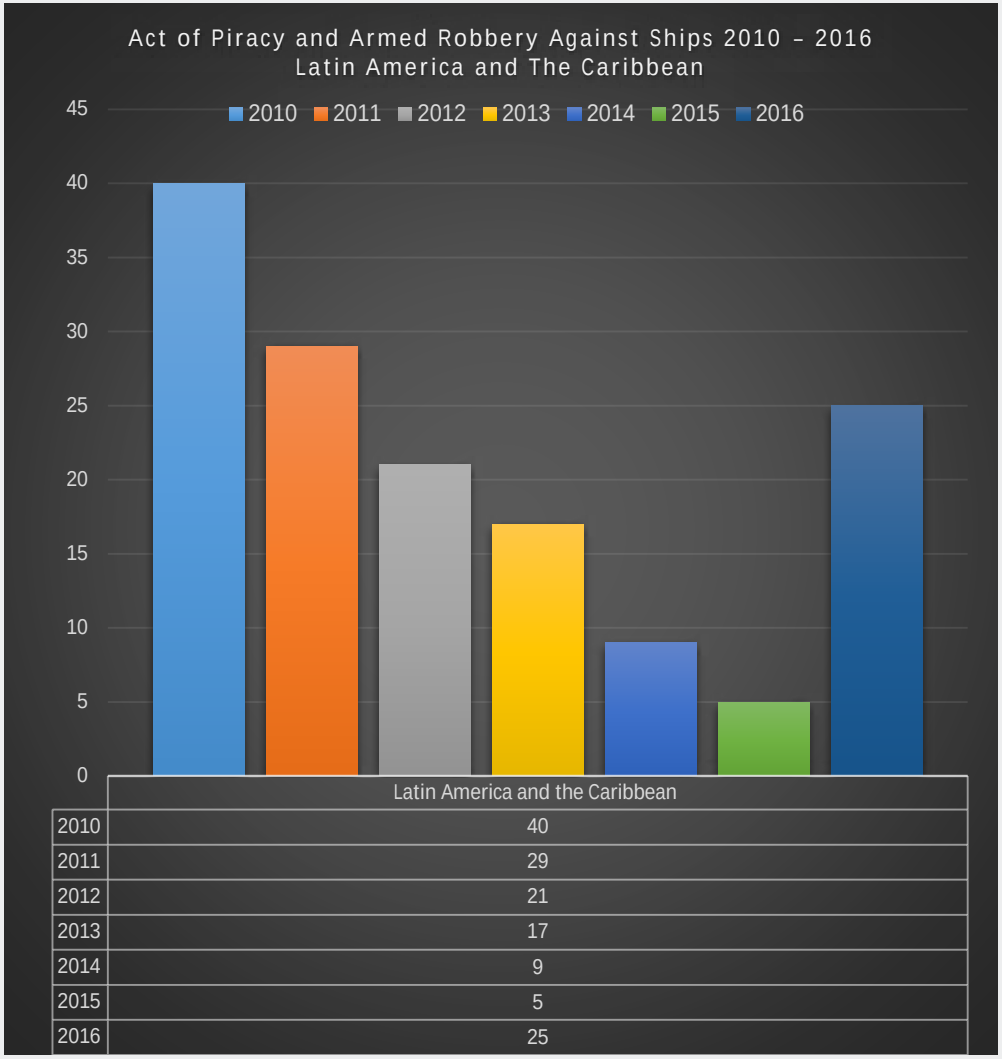
RESPOND TO SITATIONS BY USING THE APPROPRIATE TOOLS

Reports on acts of piracy and armed robbery against ships



	Malacca Strait	Indian Ocean	East Africa	West Africa	Latin America and the Caribbean	Mediterranean Sea	North Atlantic	South China Sea	Arabian Sea	Others	Total
2010	3	77	172	47	40	1		131	16	2	489
2011	22	63	223	61	29	2	2	114	28		544
2012	24	33	61	64	21	6	1	90	38	3	341
2013	17	31	20	54	17	8	1	142	6	2	298
2014	81	44	3	45	9		1	93	12	3	291
2015	134	18	4	35	5	2	2	81	15	7	303
2016	20	10	4	59	25	0	3	68	19	7	215

Reports on acts of piracy and armed robbery against ships

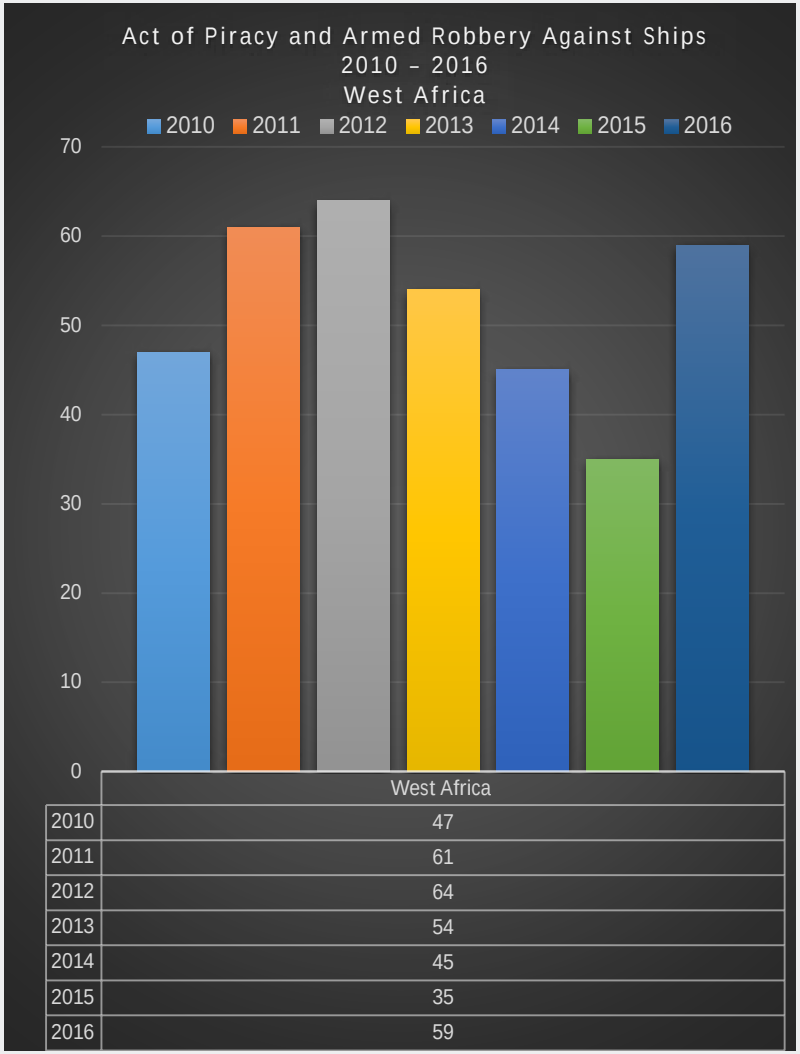


	Latin America and the Caribbean
2010	40
2011	29
2012	21
2013	17
2014	9
2015	5
2016	25





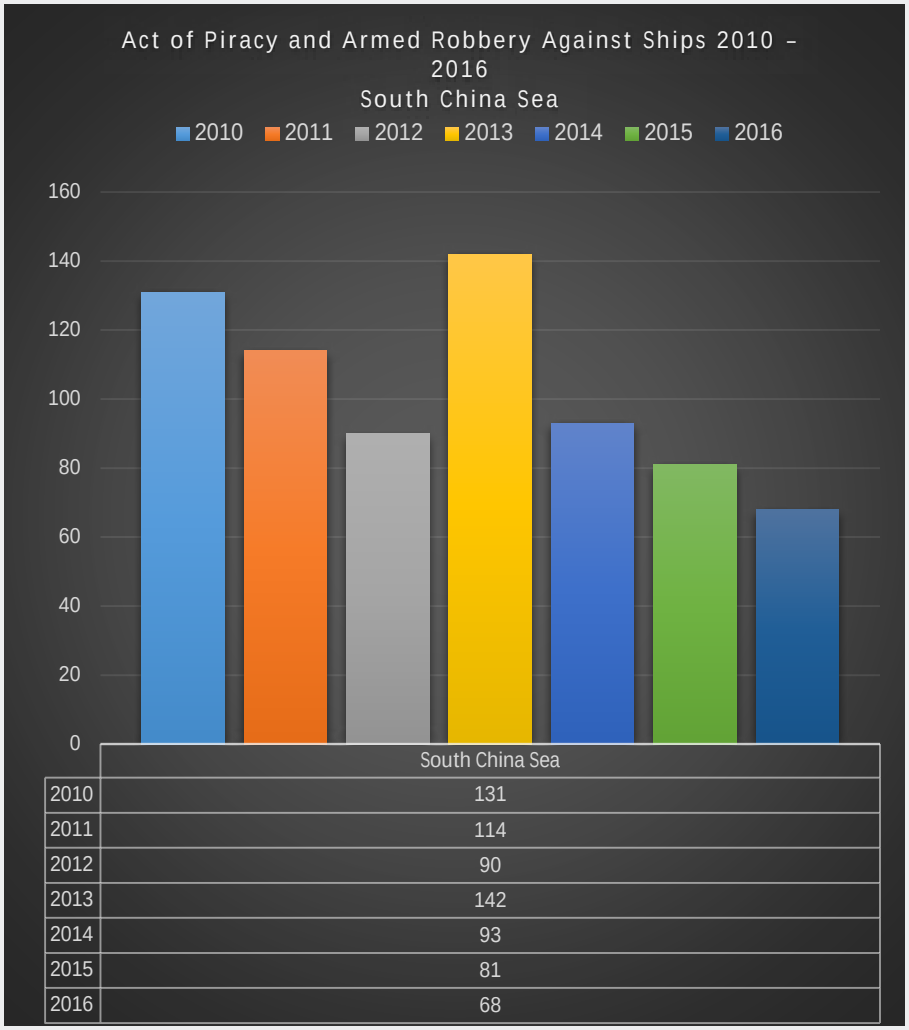
Reports on acts of piracy and armed robbery against ships



	West Africa
2010	47
2011	61
2012	64
2013	54
2014	45
2015	35
2016	59



Reports on acts of piracy and armed robbery against ships



	South China Sea
2010	131
2011	114
2012	90
2013	142
2014	93
2015	81
2016	68



Latest attack in Southern Philippines

- 19 February 2017:
- Vietnamese Bulk Carrier “MV Giang Hai”
- One dead, seven kidnapped



IMO Secretary General



Kitack Lim, Secretary General, International Maritime Organization (IMO)

IMO – World Maritime Day



28 September 2017

Technical assistance

- Needs assessment.
- Table-top excercises
- ISPS tranining for DA and PFSOs, including train-the-trainer option
- ISPS Self-asessment and audit training Workshops
- ISPS Drills and excercises Workshops
- Legal asessment, review of national legislation
- Training material for security guards with designated security duties
- Maritime Domain Awareness.



- Contact the IMO Secretariat at

marsec@imo.org

The importance of ports and port facilities



International Maritime Organization

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